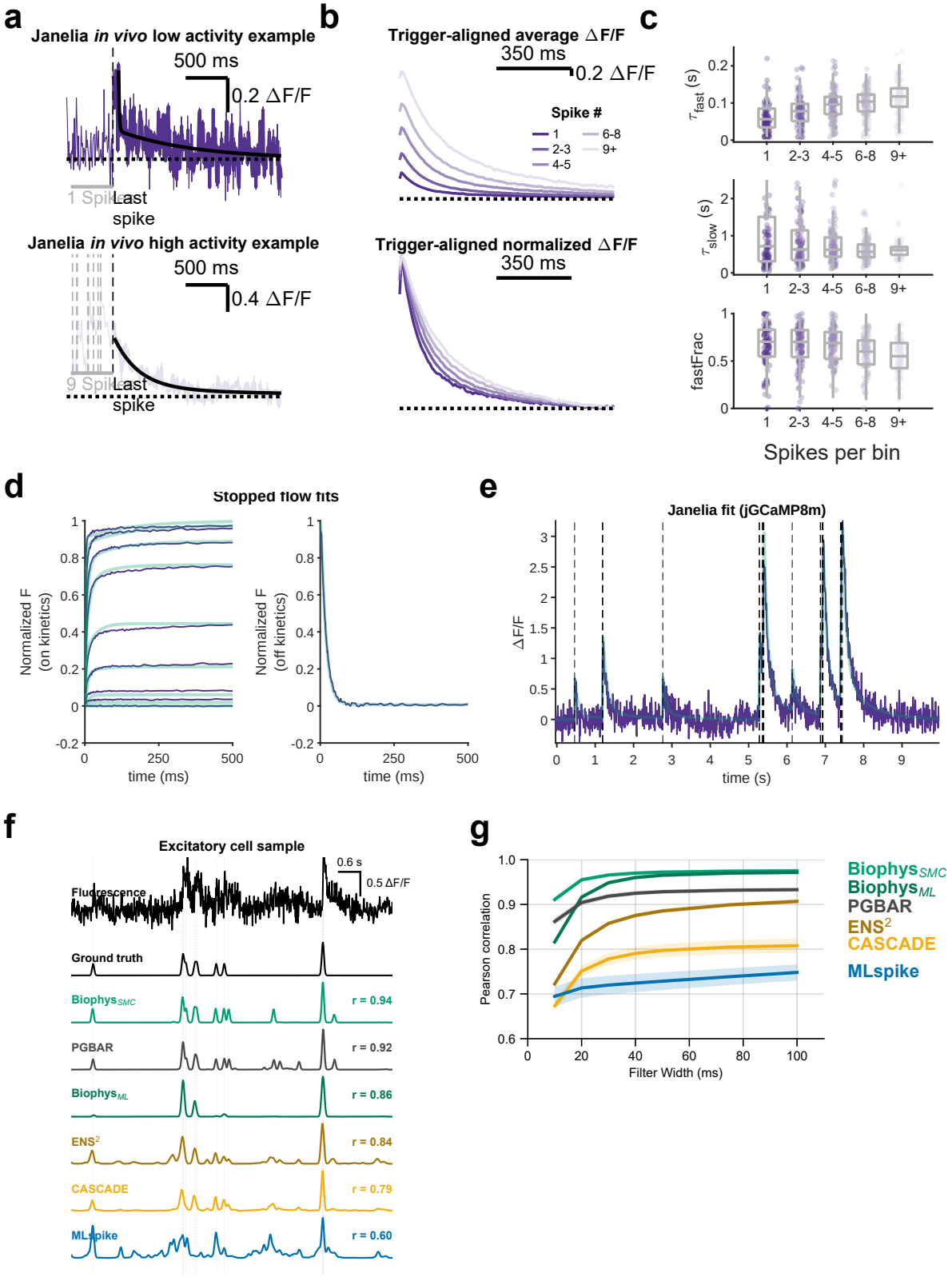

a
Janelia in vivo low activity example
500 ms
0.2 ΔF/F
1 Spike
Last spike
Janelia in vivo high activity example
500 ms
0.4 ΔF/F
9 Spikes
Last spike
b
Trigger-aligned average ΔF/F
350 ms
0.2 ΔF/F
Spike #
1
2-3
4-5
6-8
9+
Trigger-aligned normalized ΔF/F
350 ms
c
τfast (s)
τslow (s)
fastFrac
1
2-3
4-5
6-8
9+
Spikes per bin
d
Stopped flow fits
Normalized F (on kinetics)
Normalized F (off kinetics)
time (ms)
e
Janelia fit (jGCaMP8m)
ΔF/F
time (s)
f
Excitatory cell sample
0.6 s
0.5 ΔF/F
Fluorescence
Ground truth
Biophys SMC
r = 0.94
PGBAR
r = 0.92
Biophys ML
r = 0.86
ENS2
r = 0.84
CASCADE
r = 0.79
MLspike
r = 0.60
g
Pearson correlation
Filter Width (ms)
Biophys SMC
Biophys ML
PGBAR
ENS2
CASCADE
MLspike